

Subsets Cardinality and Types of Sets

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A **set** is a well-defined collection of objects called **elements**.

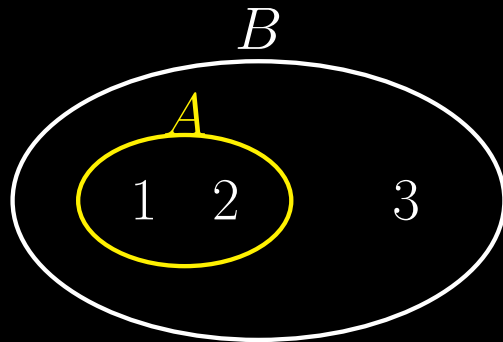
Subset

A **subset** is a set whose every element belongs to another set.

If A is a subset of B , we write

$$A \subseteq B$$

Subset Diagram



$$A = \{1, 2\} \subseteq B = \{1, 2, 3\}$$

Not a Subset

If even one element is not in the other set, it is **not** a subset.

$$\{1, 4\} \not\subseteq \{1, 2, 3\}$$

because $4 \notin \{1, 2, 3\}$.

Empty Set as a Subset

The **empty set**, denoted by $\emptyset$, is a subset of every set.

$$\emptyset \subseteq A$$

A Set as Its Own Subset

Every set is a subset of itself.

$$A \subseteq A$$

Cardinality

The **cardinality** of a set is the number of elements it contains. It is written as $n(A)$ or $|A|$.

$$A = \{a, e, i\}$$

$$n(A) = 3$$

Finite Set

A **finite set** has a limited number of elements.

$$\{1, 2, 3, 4, 5\}$$

You can count all of its elements.

Infinite Set

An **infinite set** has endlessly many elements.

$$\{1, 2, 3, \dots\}$$

The dots $\dots$ mean the pattern continues without end.

Number of Subsets

A set with n elements has

$$2^n$$

subsets.

Why 2^n ?

Each element has **2 choices**: it is either **in** the subset or **out**.

For n elements:

$$\underbrace{2 \times 2 \times \cdots \times 2}_{n \text{ times}} = 2^n$$

Listing All Subsets

List the subsets by size: 0 elements, 1 element, 2 elements, and so on.

Start with $\emptyset$. End with the set itself.

Summary of Symbols

Symbol	Meaning
$A \subseteq B$	A is a subset of B
$A \not\subseteq B$	A is not a subset of B
$\emptyset$	empty set
$n(A)$ or $ A $	cardinality of A

Examples

Example 1

List all the subsets of $R = \{r, s\}$.

Size	Subsets
0	$\emptyset$
1	$\{r\}, \{s\}$
2	$\{r, s\}$

$2^2 = 4$ subsets.

Example 2

List all the subsets of $S = \{7, 8, 9\}$.

Size	Subsets
0	$\emptyset$
1	$\{7\}, \{8\}, \{9\}$
2	$\{7, 8\}, \{7, 9\}, \{8, 9\}$
3	$\{7, 8, 9\}$

$2^3 = 8$ subsets.

Example 3

Determine the cardinality of each set.

$$A = \{3, 6, 9, 12\} \quad n(A) = 4$$

$$B = \{r, s, t, u, v\} \quad n(B) = 5$$

$$C = \{0\} \quad n(C) = 1$$

Example 4

Tell whether each set is finite or infinite.

(a) $\{5, 10, 15, \dots\}$ **Infinite**

(b) The set of letters in the English alphabet.
Finite

(c) The set of multiples of 7. **Infinite**

Example 5

Let $F = \{1, 3, 5, 7\}$. Is each a subset of F ?

(a) $\{3, 7\}$ Yes

(b) $\{1, 4\}$ No, because $4 \notin F$

(c) $\emptyset$ Yes

(d) F Yes

Example 6

Without listing, find the number of subsets of $G = \{k, l, m, n, o\}$.

$$n(G) = 5$$

$$2^5 = 32$$

G has 32 subsets.

Practice Exercises

Practice 1

List all the subsets of $A = \{1, 2\}$.

Practice 2

List all the subsets of $B = \{a, b, c\}$.

Practice 3

List all the subsets of $C = \{x, y, z, w\}$.

Practice 4

Determine the cardinality of each set.

(a) $\{5, 10, 15, 20\}$

(b) $\{a, e, i, o, u\}$

(c) $\emptyset$

Practice 5

Determine the cardinality of each set.

(a) $\{2, 4, 6, 8, 10, 12\}$

(b) $\{m, n, p\}$

(c) $\{100\}$

Practice 6

Tell whether each set is finite or infinite.

(a) $\{1, 2, 3, \dots\}$

(b) $\{5, 10, 15\}$

(c) The set of vowels in the English alphabet.

Practice 7

Tell whether each set is finite or infinite.

- (a) The set of all whole numbers.
- (b) The set of months in a year.
- (c) The set of even numbers.

Practice 8

Let $D = \{a, b, c\}$.

- (a) Is $\{a, c\}$ a subset of D ?
- (b) Is $\{a, d\}$ a subset of D ?
- (c) Is $\emptyset$ a subset of D ?

Practice 9

Let $E = \{2, 4, 6, 8\}$.

- (a) Is $\{4, 8\}$ a subset of E ?
- (b) Is $\{2, 5\}$ a subset of E ?
- (c) Is E a subset of itself?

Practice 10

Without listing all the subsets, determine the number of subsets of each set.

(a) $\{a, b\}$

(b) $\{1, 2, 3\}$

(c) $\{m, n, p, q\}$

Activities

Activity 1

List all the subsets of $A = \{3, 5\}$.

Activity 2

List all the subsets of $B = \{p, q, r\}$.

Activity 3

List all the subsets of $C = \{2, 4, 6\}$.

Activity 4

Determine the cardinality of each set.

(a) $\{7, 14, 21\}$

(b) $\{b, c, d, e, f, g\}$

(c) $\emptyset$

Activity 5

Determine the cardinality of each set.

(a) $\{11, 22, 33, 44, 55\}$

(b) $\{x, y\}$

(c) $\{50\}$

Activity 6

Tell whether each set is finite or infinite.

(a) $\{2, 4, 6, 8, \dots\}$

(b) $\{8, 16, 24, 32\}$

(c) The set of days in a week.

Activity 7

Tell whether each set is finite or infinite.

- (a) The set of integers.
- (b) The set of planets in the Solar System.
- (c) The set of odd numbers.

Activity 8

Let $D = \{m, n, o\}$.

- (a) Is $\{m, o\}$ a subset of D ?
- (b) Is $\{m, p\}$ a subset of D ?
- (c) Is $\emptyset$ a subset of D ?

Activity 9

Let $E = \{10, 20, 30, 40\}$.

- (a) Is $\{20, 40\}$ a subset of E ?
- (b) Is $\{10, 25\}$ a subset of E ?
- (c) Is E a subset of itself?

Activity 10

Without listing all the subsets, determine the number of subsets of each set.

(a) $\{x, y\}$

(b) $\{4, 5, 6\}$

(c) $\{a, b, c, d\}$

Thank you!